

Fake News Detection Using BERT-Based Transformer Model

Ayush Kumar Sahoo

Roll number: 21, Branch: CEN

Subject: Advance Machine Learning

Department of Computer Science and Engineering

Silicon University

cen.23bcd80@silicon.ac.in

Abstract

The rapid spread of misinformation across digital platforms has made fake news detection a critical challenge. This work proposes a transformer-based approach using BERT for classifying news articles as fake or real. The model leverages contextual embeddings and attention mechanisms to capture semantic relationships in text. Experimental evaluation on a real-world dataset demonstrates an accuracy of 82% with balanced precision, recall, and F1-score. The results highlight the superiority of transformer-based models over traditional machine learning approaches.

I. Introduction

With the rise of social media and digital communication, fake news has become a major concern. Misinformation can influence elections, public opinion, and societal behavior. Traditional verification methods are slow and inefficient. Automated fake news detection using Natural Language Processing (NLP) has emerged as a viable solution. Transformer-based architectures, particularly BERT, have revolutionized NLP by enabling deep contextual understanding.

II. Related Work

Earlier approaches relied on machine learning techniques such as Naive Bayes and Support Vector Machines. These methods depend on handcrafted features and fail to capture context. Deep learning models such as LSTM improved performance but struggled with long-range dependencies. Transformer models overcome these limitations

using attention mechanisms, making them highly effective for text classification.

III. Methodology

A. Dataset

The dataset consists of fake and real news articles obtained from Kaggle. The dataset was cleaned, labeled, and reduced to approximately 3000 samples for efficient training.

B. Data Preprocessing

Preprocessing steps include removing URLs, converting text to lowercase, removing special characters, and tokenizing text using the BERT tokenizer. These steps ensure clean and consistent input data.

C. Model Architecture

The proposed system uses BERT, a transformer-based architecture that processes text bidirectionally. It uses multi-head self-attention to understand relationships between words. A classification layer is added on top for binary classification.

IV. Experimental Results

The model achieved strong performance on the validation dataset. Key metrics are summarized below:

Accuracy: 82%

Precision: 0.83

Recall: 0.82

F1 Score: 0.82

A. Training Performance

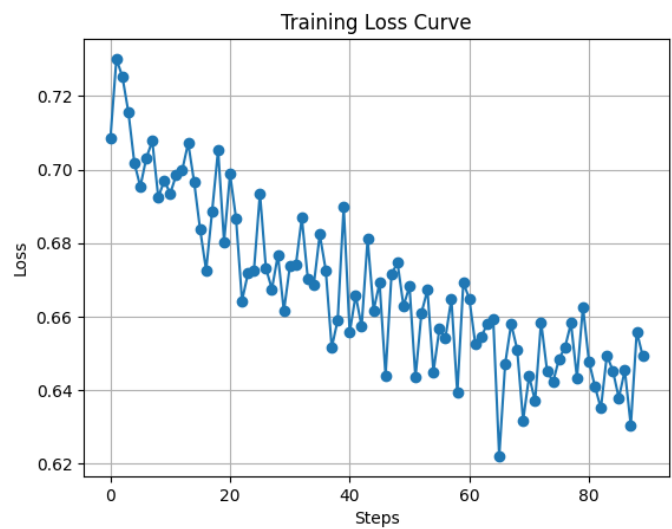


Fig. 1. Training Loss Curve

B. Confusion Matrix Analysis

The confusion matrix illustrates the classification performance of the model. Most predictions lie along the diagonal, indicating correct classification. Some misclassifications exist, particularly between fake and real categories, but overall performance is strong.

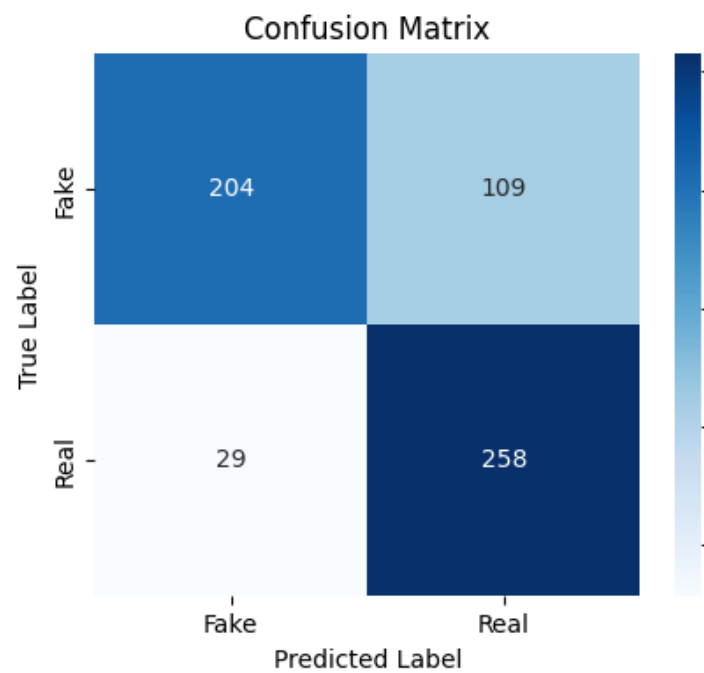
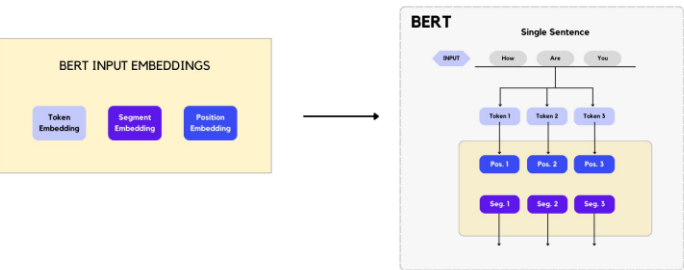


Fig. 2. Confusion Matrix

The training loss decreased steadily, indicating proper convergence of the model.



V. Discussion

The results demonstrate that BERT effectively captures contextual relationships in text. The use of class weights improved model performance. Compared to traditional models, BERT provides higher accuracy and robustness.

VI. Conclusion

This study presents a BERT-based fake news detection system achieving 82% accuracy. The results confirm the effectiveness of transformer-based models for NLP tasks.

VII. Future Work

Future work includes using larger datasets, fine-tuning advanced models such as RoBERTa, and deploying the system as a web application.

References

- [1] Y. LeCun et al., Deep Learning, 2015
- [2] J. Devlin et al., BERT, 2018
- [3] HuggingFace Documentation
- [4] Kaggle Dataset